



East West University

Dept. of Mathematical and Physical Sciences (MPS)

Assignment-01, Fall: 2025, (Section 8)

Course Instructor: Dr. Md. Shariful Alam

Course No. MAT205 Course Title: Linear Algebra and Complex Variables

Instructions:

- Prepare the question on the following Topics and solve them individually by hand, Scan and name the file as ID_Name.
- Write your name, give signature, submission date and ID in each page of the Assignment-01.
- Submit your Assignment-01 (**scanned pdf copy**) through Google classroom.
- **The last date of submission of your Assignment-01 is 23 November (Sunday), 2025**

N. B. The evaluation procedure depends on the Choice of the question pattern for each topic and their details solution method.

Assignment Topics:

1. Matrix and Solution of System of Linear equations.
2. Vector Space.
3. Linear Transformation.
4. Eigenvalues and Eigenvectors including Cayley-Hamilton Theorem.
5. Diagonalization of a Matrix and Inner Product Space.